LAB 2 REPORT

CYOP 300 - BUILDING SECURE WEB APPLICATIONS

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Executive Summary

Attached is my submission for lab 2. While no particular difficulties were encountered, there are a few items that I would like to discuss in my submission.

First, I included type hints throughout the project. In day-to-day work, I find that type hints help resolve questions before they turn into problems.

Second, I split out the sections of code that were more complicated from the main main.py code. These include a common methods package, the prompt user for input code, as well as the password generation code.

Third, I used a python module to handle presenting a menu and parsing user input.

For the calculation of the number of days from today to July 4th, 2025, I chose to present the text indicating that the date was in the past, formatted that it was *X* days ago. The code handles both dates that occur in the past, as well as future dates.

For the volume of the cylinder, I prompt the user for the numerical value, as well as the (metric) units. The result, however, is presented only in mm^3.

Full unit tests were generated, and are included in the project zip file.

Testing

Test Case Table

Test Cases	Description	Expected Output	Actual Output	Pass/Fail
Test Case 1	Generate password of: Length -1 Upper y Lower y Digits y Special y	Invalid password length	Invalid input: Password length is too short for the selected complexity.	pass
Test Case 2	Generate password of: Length 128 Upper y Lower y Digits y Special y	Randomly generated password of desired length	Your secure password is: 4,5;Q<0Ycen.fDc:G7Yk!1{G(6U&w!2^E]uMwrglw=.6@{o&f&aJq,2si\]+9.hssYgt2mwiEmb3(!4'8F8y{Ye7x"21rb@BD>,Wt`R:;C5%~B1K0si`kD.RQ6+]ragg Your password consists of: 24 upper characters 42 lower characters 22 digits characters 40 special characters	
Test Case 3	Generate password of: Length 1025 Upper y Lower y Digits y Special y	Randomly generated password of desired length	Your secure password is: JzJv3xz853>YS%$f;)Y'%K%0(F0,eyfbsz%1+w222CTENK\|z5wU"vdqggpuju-\|ei[{a;'<9~JwV=y9nc}?fOhKp;QW"]99$hur+>"bL8'dcg+pZ^&*&cd7Yx;#aEi',.p*g?1;k7%:KDqQCu^_q.jFkb|H|;@*,1%[3=!k{;+/)p"s6rM7._?EPuYm&#Rt%nmAEZg0=v0lZpr|}!&YV,/Ix6Y:R;aiJ_pOg&#?2y8c9o_nwt[Q2u#lVW+tu*<5<]7&=XKBzB%KrK7+s&xE?*E(n#F]'F{Aa%Ie_Z`^A]=#}]o[)g:T'~L&Z*@n;"n|hE}]PeRsYTX*XNL#$!;[p<KUS,qoF^1:NJzT#wKY=>36##4S:U=~+syHg\|#OHdhhJ]?2C#<0YC.wf\|.f"_f_Z~?P7_2AP-$~jWEeV0MMqnv~wv-MuN;y5doyc<~JGgaooJ6	

			G8U@o;?a- oxU\q=dS4CYkaPAe>Y+?? GeDOwuRUP)#mP`H~[& Qf{jL0t<cfV%OzT=H>(QS8 pRQ=**#c*ctY_^SrHmnU &0YCBusq"_wVw},O;,sKb AU.8gLySMZ4&O@y@EEl 6g;YVVB)xl*4,%J'Oy!wOo mj#ZM[3RtYS@f]u<CO1% tM)Qo/i[D%|L6uB.[tP*]D $z"9t$UJQJy;h.+}u%~r%] ^FTgnMr(doy7Z~?|twCOI A:$&_{=M;s.#"iS>Uclt,99. qu6kbqfk+r2nCKX=j0V]_J ",ta(bge=lJZfzm_Z)WG:{P <c+7Egy\AV.%L;WJv&^izP ypvAxw\YD9<?!e%**z?k A5b^AbdRFf3^r@pj`ti!&h d[g)9:5[*//KhM,_5<fu}vR fcRq@~_Z>Zk/QZnp^Z3q RqFc77k?TB~^JGNS\:3+u +B+pm}rq15ePwG~Yww# u!VER7!KV/o[=IC:_,BPb^^ ;I$1C6>S%%xd1=3A`Dg/q fBk\9f>>*D>AGWrH\4?2J eYl),%Kk:mDw>a%WL~L< z>#x;|=;P:D,s*`mWu5r[#g *h|{bW7.JhL;n Your password consists of: 267 upper characters 298 lower characters 86 digits characters 374 special characters	
Test Case 4	Generate password of: Length 20 Upper y Lower n Digits n Special n	Randomly generated password of desired length	Your secure password is: GSZKXJXLHYYAOEQGEAE Q Your password consists of: 20 upper characters 0 lower characters 0 digits characters 0 special characters	
Test Case 5	Generate password of: Length 20 Upper y	Randomly generated password of desired length	Your secure password is: I/_.|@A?F;|4<JG5H/$I	

	Lower n Digits y Special y		Your password consists of: 7 upper characters 0 lower characters 2 digits characters 11 special characters	
Test Case 6	Calculate percentage: Numerator 59 Denominator 12 Decimal 4	59/12=4.9167, so, 491.6667%	59.0/12.0 = 491.6667 %	
Test Case 7	Calculate percentage: Numerator 0 Denominator 12 Decimal -4	Error message that the number of decimal spaces cannot be negative	Error: Decimal places cannot be negative.	
Test Case 8	Calculate percentage: Numerator 59 Denominator 0 Decimal 4	Error message that the denominator cannot be zero	Error: Decimal places cannot be negative.	
Test Case 9	Days from 7/4/25	263 days ago	July 4, 2025 occurred 263 days ago.	
Test Case 10	Calculate triangle: Leg 1: 15 Leg 2: 45 Angle: 50	$\sqrt{a^2 + b^2 - (2ab\cos C)}$ = 37.18	The length of the leg of the triangle is: 37.18	
Test Case 11	Calculate triangle: Leg 1: 45 Leg 2: 45 Angle: 90	$\sqrt{a^2 + b^2 - (2ab\cos C)}$ = 63.64	The length of the leg of the triangle is: 63.64	
Test Case 12	Calculate triangle: Leg 1: no value provided	Error message that a value is required.	Invalid input. Please enter a valid float.	
Test Case 13	Volume of right cylinder: Radius: 1m Height: 39cm	Calculated volume at 12.25 mm^3	The volume of the cylinder is: 12.25 mm^3	
Test Case 14	Volume of right cylinder: Radius: 1km Height: 39mm	Calculated volume at 1,225,221.13 mm^3	The volume of the cylinder is: 1225221.13 mm^3	
Test Case 15	Exit the program	Program exits with a thank you message.		

Test Case Screenshots

Test Case 1

```
Please enter the length of the password (Integers only): -1
Please enter the complexity of the password via the following prompts:
Include uppercase letters? (y/n): y
Include lowercase letters? (y/n): y
Include digits? (y/n): y
Include special characters? (y/n): y

Invalid input: Password length is too short for the selected complexity.

Please try again.
```

Test Case 2

```
Please enter the length of the password (Integers only): 128
Please enter the complexity of the password via the following prompts:
Include uppercase letters? (y/n): y
Include lowercase letters? (y/n): y
Include digits? (y/n): y
Include special characters? (y/n): y
Your secure password is: 4,5;Q<0Ycen.fDc:G7Yk!1{G(6U&w!2^E]uMwrglw=.6@{o&f&aJq,2si\]+9.hssYgt2mwiEmb3(!4'8F8y{Ye7x"21rb@BD>,Wt`R:;C5%~B1K0si`kD.RQ6+]ragg

Your password consists of:
24 upper characters
42 lower characters
22 digits characters
40 special characters
> Generate a new Secure Password
  Return to Main Menu
  Exit
```

Test Case 3

```
Please enter the length of the password (Integers only): 1025
Please enter the complexity of the password via the following prompts:
Include uppercase letters? (y/n): y
Include lowercase letters? (y/n): y
Include digits? (y/n): y
Include special characters? (y/n): y
Your secure password is: JzJv3xz853>YS%$f;)Y'%K%0(F0,eyfbsz%1+w222CTENK|z5wU"vdqggpuju-|ei[{a;'<9~JwV=y9nc}?fOhKp;QW"]99$hur+>"bL8'dcg+pZ^&*&cd7Yx;#aEi',.p*g?1;k7%:KDqQCu^_q.jFkb|H|;@*,1%[3=!k{;+/
)p"s6rM7._?EPuYm&#Rt%nmAEZg0=v0lZpr|}!&YV,/Ix6Y:R;aiJ_pOg&#?2y8c9o_nwt[Q2u#lVW+tu*<5<]7&=XKBzB%KrK7+s&xE?*E(n#F]'F{Aa%Ie_Z`^A]=#}]o[]g:T'~L&Z*@n;"n|hE}]PeRsYTX*XNL#$!;[p<KUS,qoF^1:NJzT#wKY=>36##4S
:U=~+syHg|#OHdhhJ]?2C#<0YC.wf\|.f"_f_Z~?P7_2AP-$~jWEeV0MMqnv~wv-MuN;y5doyc<~J6gaooJ6G8U@o;?a-oxU\q=dS4CYkaPAe>Y+??GeDOwuRUP)#mP`H~[&Qf{jL0t<cfV%OzT=H>(QS8pRQ=**#c*ctY_^SrHmnU&0YCBusq"_wVw},0;,sKbA
U.8gLySMZ4&O@y@EEL6g;YVVB)xl*4,%J'Oy!wOomj#ZM[3RtYS@f]u<CO1%tM)Qo/i[D%|L6uB.[tP*]D$z"9t$UJQJy;h.+}u%~r%]^FTgnMr(doy7Z~?|twCOIA:$&_{=M;s.#"iS>Uclt,99.qu6kbqfk+r2nCKX=j0V]_J",ta(bge=lJZfzm_Z)WG:{P<c
+7Egy\AV.%L;WJv&^izPypvAxw\YD9<?!e%**z?kA5b^AbdRFf3^r@pj`ti!&hd[g)9:5[*//KhM,_5<fu}vRfcRq@~_Z>Zk/QZnp^Z3qRqFc77k?TB~^J6NS\:3+u+B+pm}rq15ePwG~Yww#u!VER7!KV/o[=IC:_,BPb^^;I$1C6>S%%xd1=3A`Dg/qfBk\9f>
>*D>AGWrH\4?2JeYl),%Kk:mDw>a%WL~L<z>#x;|=;P:D,s*`mWu5r[#g*h|{bW7.JhL;n

Your password consists of:
267 upper characters
298 lower characters
86 digits characters
374 special characters
> Generate a new Secure Password
  Return to Main Menu
```

Test Case 4

```
Please enter the length of the password (Integers only): 20
Please enter the complexity of the password via the following prompts:
Include uppercase letters? (y/n): y
Include lowercase letters? (y/n): n
Include digits? (y/n): n
Include special characters? (y/n): n
Your secure password is: GSZKXJXLHYYAOEQGEAEQ

Your password consists of:
20 upper characters
0 lower characters
0 digits characters
0 special characters
```

Test Case 5

```
Please enter the length of the password (Integers only): 20
Please enter the complexity of the password via the following prompts:
Include uppercase letters? (y/n): y
Include lowercase letters? (y/n): n
Include digits? (y/n): y
Include special characters? (y/n): y
Your secure password is: I/_.|@A?F;|4<JG5H/$I

Your password consists of:
7 upper characters
0 lower characters
2 digits characters
11 special characters
```

Test Case 6

```
Enter the numerator, denominator, and decimal places:
Numerator: 59
Denominator: 12
Decimal Places: 4
59.0/12.0 = 491.6667 %
```

Test Case 7

```
Enter the numerator, denominator, and decimal places:
Numerator: 0
Denominator: 12
Decimal Places: -4
Error: Decimal places cannot be negative.
```

Test Case 8

```
Enter the numerator, denominator, and decimal places:
Numerator: 59
Denominator: 0
Decimal Places: 4
Error: Denominator cannot be zero.
```

Test Case 9

```
July 4, 2025 occurred 263 days ago.
```

Test Case 10

```
Enter the lengths of the sides of the triangle:
Side a: 15
Side b: 45
Angle C (in degrees): 50
The length of the leg of the triangle is: 37.18
```

Test Case 11

```
Enter the lengths of the sides of the triangle:
Side a: 45
Side b: 45
Angle C (in degrees): 90
The length of the leg of the triangle is: 63.64
```

Test Case 12

```
Enter the lengths of the sides of the triangle:
Side a:
Invalid input. Please enter a valid float.
Side a:
```

Test Case 13

```
Enter the radius and height of the cylinder:
Please enter the radius of the circle: 1
Please enter the unit of the radius (km, m, cm, mm): m
Please enter the height of the cylinder: 39
Please enter the unit of the height (km, m, cm, mm): cm
The volume of the cylinder is: 12.25 mm^3
```

Test Case 14

```
Enter the radius and height of the cylinder:
Please enter the radius of the circle: 1
Please enter the unit of the radius (km, m, cm, mm): km
Please enter the height of the cylinder: 39
Please enter the unit of the height (km, m, cm, mm): mm
The volume of the cylinder is: 1225221.13 mm^3
```

Test Case 15

```
Exiting program. Thank you for using me!
```

Pylint Results

```
************* Module lab2.lab2prompt
lab2prompt.py:10:0: C0301: Line too long (155/100) (line-too-long)
************* Module lab2.main
main.py:41:4: W0105: String statement has no effect (pointless-string-
statement)
main.py:80:4: W0212: Access to a protected member _Prompt of a client
class (protected-access)
main.py:114:4: W0212: Access to a protected member _Prompt of a client
class (protected-access)
main.py:147:4: W0212: Access to a protected member _Prompt of a client
class (protected-access)
main.py:185:4: W0212: Access to a protected member _Prompt of a client
class (protected-access)
main.py:214:4: W0212: Access to a protected member _Prompt of a client
class (protected-access)
main.py:200:12: W0612: Unused variable 'radius_unit' (unused-variable)
main.py:201:12: W0612: Unused variable 'height_unit' (unused-variable)
main.py:252:4: W0212: Access to a protected member _Prompt of a client
class (protected-access)
************* Module lab2.lab2passwd
lab2passwd.py:10:0: C0301: Line too long (155/100) (line-too-long)
lab2passwd.py:24:0: C0301: Line too long (101/100) (line-too-long)
************* Module lab2.lab2common
lab2common.py:10:0: C0301: Line too long (155/100) (line-too-long)
lab2common.py:54:0: C0301: Line too long (105/100) (line-too-long)
lab2common.py:55:0: C0301: Line too long (107/100) (line-too-long)
lab2common.py:57:0: C0301: Line too long (104/100) (line-too-long)
lab2common.py:61:0: C0301: Line too long (104/100) (line-too-long)
lab2common.py:70:8: R1723: Unnecessary "elif" after "break", remove the
leading "el" from "elif" (no-else-break)
************* Module lab2.tests.test_lab2passwd
tests/test_lab2passwd.py:1:0: C0114: Missing module docstring (missing-
module-docstring)
tests/test_lab2passwd.py:10:0: C0116: Missing function or method docstring
(missing-function-docstring)
tests/test_lab2passwd.py:27:0: C0116: Missing function or method docstring
(missing-function-docstring)
tests/test_lab2passwd.py:41:0: C0116: Missing function or method docstring
(missing-function-docstring)
tests/test_lab2passwd.py:48:0: C0116: Missing function or method docstring
(missing-function-docstring)
************* Module lab2.tests.conftest
tests/conftest.py:1:0: C0114: Missing module docstring (missing-module-
docstring)
tests/conftest.py:10:4: C0116: Missing function or method docstring
```

```
(missing-function-docstring)
tests/conftest.py:5:0: R0903: Too few public methods (1/2) (too-few-
public-methods)
************* Module lab2.tests.test_lab2common
tests/test_lab2common.py:1:0: C0114: Missing module docstring (missing-
module-docstring)
tests/test_lab2common.py:8:0: C0116: Missing function or method docstring
(missing-function-docstring)
tests/test_lab2common.py:18:0: C0116: Missing function or method docstring
(missing-function-docstring)
tests/test_lab2common.py:28:0: C0116: Missing function or method docstring
(missing-function-docstring)
tests/test_lab2common.py:39:0: C0116: Missing function or method docstring
(missing-function-docstring)
************* Module lab2.tests
tests/__init__.py:12:0: C0115: Missing class docstring (missing-class-
docstring)
tests/__init__.py:13:4: C0116: Missing function or method docstring
(missing-function-docstring)
************* Module lab2.tests.test_lab2prompt
tests/test_lab2prompt.py:1:0: C0114: Missing module docstring (missing-
module-docstring)
tests/test_lab2prompt.py:6:0: C0116: Missing function or method docstring
(missing-function-docstring)
tests/test_lab2prompt.py:7:4: C0115: Missing class docstring (missing-
class-docstring)
tests/test_lab2prompt.py:11:8: C0116: Missing function or method docstring
(missing-function-docstring)
tests/test_lab2prompt.py:7:4: R0903: Too few public methods (1/2) (too-
few-public-methods)
tests/test_lab2prompt.py:16:13: W0212: Access to a protected member
_Prompt of a client class (protected-access)
tests/test_lab2prompt.py:21:0: C0116: Missing function or method docstring
(missing-function-docstring)
tests/test_lab2prompt.py:24:4: C0115: Missing class docstring (missing-
class-docstring)
tests/test_lab2prompt.py:28:8: C0116: Missing function or method docstring
(missing-function-docstring)
tests/test_lab2prompt.py:24:4: R0903: Too few public methods (1/2) (too-
few-public-methods)
tests/test_lab2prompt.py:33:13: W0212: Access to a protected member
_Prompt of a client class (protected-access)
tests/test_lab2prompt.py:39:0: C0116: Missing function or method docstring
(missing-function-docstring)
tests/test_lab2prompt.py:46:8: W0212: Access to a protected member _Prompt
of a client class (protected-access)
```

```
tests/test_lab2prompt.py:49:4: W0212: Access to a protected member _Prompt
of a client class (protected-access)
tests/test_lab2prompt.py:54:0: C0116: Missing function or method docstring
(missing-function-docstring)
tests/test_lab2prompt.py:61:24: W0212: Access to a protected member
_Prompt of a client class (protected-access)
tests/test_lab2prompt.py:71:4: W0212: Access to a protected member _Prompt
of a client class (protected-access)
tests/test_lab2prompt.py:1:0: W0611: Unused import pytest (unused-import)
************* Module lab2.tests.test_main
tests/test_main.py:1:0: C0114: Missing module docstring (missing-module-
docstring)
tests/test_main.py:10:0: C0116: Missing function or method docstring
(missing-function-docstring)
tests/test_main.py:23:8: W0212: Access to a protected member _Prompt of a
client class (protected-access)
tests/test_main.py:36:0: C0116: Missing function or method docstring
(missing-function-docstring)
tests/test_main.py:43:8: W0212: Access to a protected member _Prompt of a
client class (protected-access)
tests/test_main.py:53:0: C0116: Missing function or method docstring
(missing-function-docstring)
tests/test_main.py:60:8: W0212: Access to a protected member _Prompt of a
client class (protected-access)
tests/test_main.py:70:0: C0116: Missing function or method docstring
(missing-function-docstring)
tests/test_main.py:71:4: C0115: Missing class docstring (missing-class-
docstring)
tests/test_main.py:78:8: W0212: Access to a protected member _Prompt of a
client class (protected-access)
tests/test_main.py:87:0: C0116: Missing function or method docstring
(missing-function-docstring)
tests/test_main.py:88:4: C0115: Missing class docstring (missing-class-
docstring)
tests/test_main.py:95:8: W0212: Access to a protected member _Prompt of a
client class (protected-access)
tests/test_main.py:104:0: C0116: Missing function or method docstring
(missing-function-docstring)
tests/test_main.py:111:8: W0212: Access to a protected member _Prompt of a
client class (protected-access)
tests/test_main.py:121:0: C0116: Missing function or method docstring
(missing-function-docstring)
tests/test_main.py:126:8: W0212: Access to a protected member _Prompt of a
client class (protected-access)
tests/test_main.py:137:0: C0116: Missing function or method docstring
(missing-function-docstring)
```

```
tests/test_main.py:137:41: W0613: Unused argument 'monkeypatch' (unused-
argument)
tests/test_main.py:146:0: C0116: Missing function or method docstring
(missing-function-docstring)
tests/test_main.py:153:8: W0212: Access to a protected member _Prompt of a
client class (protected-access)
tests/test_main.py:3:0: W0611: Unused import sys (unused-import)

------------------------------------
Your code has been rated at 8.05/10

Run 2

************* Module lab2.lab2common
lab2common.py:10:0: C0301: Line too long (155/100) (line-too-long)
lab2common.py:54:0: C0301: Line too long (105/100) (line-too-long)
lab2common.py:55:0: C0301: Line too long (107/100) (line-too-long)
lab2common.py:57:0: C0301: Line too long (104/100) (line-too-long)
lab2common.py:61:0: C0301: Line too long (104/100) (line-too-long)
lab2common.py:70:8: R1723: Unnecessary "elif" after "break", remove the
leading "el" from "elif" (no-else-break)
************* Module lab2.lab2passwd
lab2passwd.py:10:0: C0301: Line too long (155/100) (line-too-long)
lab2passwd.py:24:0: C0301: Line too long (101/100) (line-too-long)
************* Module lab2.lab2prompt
lab2prompt.py:15:0: E0401: Unable to import 'simple_term_menu' (import-
error)
************* Module lab2.main
main.py:200:12: W0612: Unused variable 'radius_unit' (unused-variable)
main.py:201:12: W0612: Unused variable 'height_unit' (unused-variable)

------------------------------------
Your code has been rated at 9.14/10

Run 3

************* Module lab2.lab2common
lab2common.py:82:8: R1723: Unnecessary "else" after "break", remove the
"else" and de-indent the code inside it (no-else-break)
************* Module lab2.main
main.py:200:12: W0612: Unused variable 'radius_unit' (unused-variable)
main.py:201:12: W0612: Unused variable 'height_unit' (unused-variable)

--------------------------------------------------------------------
Your code has been rated at 9.83/10 (previous run: 9.14/10, +0.69)
```

```
Run 4
************* Module lab2.main
main.py:200:12: W0612: Unused variable 'radius_unit' (unused-variable)
main.py:201:12: W0612: Unused variable 'height_unit' (unused-variable)

------------------------------------------------------------------
Your code has been rated at 9.89/10 (previous run: 9.83/10, +0.06)
```